

Linear Algebra I Recitation Class: Week 4

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Throughout these notes, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ denotes the sets of rational, real, and complex numbers, respectively. V is a vector space over a field F . If you are not yet familiar with fields, you may keep the examples $F = \mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$ in mind.

A subset S of V is called a **generating set** for V if $\text{span}(S) = V$. An **F -basis** for V is a linearly independent generating set β of V . If V has an F -basis which is finite, then V is called **finite dimensional**. Otherwise, it is called **infinite dimensional**. We will first focus on the case in which V is finite dimensional. We recall some important results in this section:

Theorem 1 (Theorem 1.9). *Let S be a finite generating set of V , then some subsets of S are bases for V .*

$\implies$ We can obtain bases for V by reducing a finite generating set for V .

Theorem 2 (Theorem 1.10 -Replacement Theorem-). *Let S be a generating set of V with $|S| = n$ and let T be a linearly independent subset of V with $|T| = m$. Then $m \leq n$ and there exists a subset γ of S with $|\gamma| = n - m$ such that $T \cup \gamma$ generates V .*

$\implies$ We can extend any linear independent set of V to a generating set of V .

Theorem 3 (Corollary 1). *Every basis for V contains the same number of vectors.*

$\implies$ We can define the **dimension** $\dim_F V$ of V (over F) to be the number of a basis for V .

Theorem 4 (Corollary 2). *Let V be an n dimensional vector space.*

1. *Any finite generating set for V contains at least n vectors, and a generating set for V that contains exactly n vectors is a basis for V .*
2. *Any linearly independent subset of V that contains exactly n vectors is a basis for V .*
3. *Every linearly independent subset of V can be extended to a basis for V .*

Theorem 5 (Theorem 1.11). *Let W be a subspace of finite dimensional vector space V , then W is also finite dimensional and $\dim W \leq \dim V$. Moreover, if $\dim W = \dim V$ then $W = V$.*

Theorem 1 can be generalized to the case where the generating set of V is infinite:

Proposition 6 (Exercise 1.6.20(a)). *Let V be a vector space having dimension n , and let S be a generating set of V , then there exists a subset of S that is a basis for V .*

Sketch. If $V = \{0\}$, $V, \emptyset$ generates V , and $\emptyset$ is a basis for V . If V is not the zero vector space, then S contains nonzero element, say $u_1 \in S$, and $\{u_1\}$ is a linearly independent set. We keep finding new elements $u_{k+1} \in S$ such that $u_{k+1} \notin \text{span}\{u_1, \dots, u_k\}$ and add in to the set. This process will end before $k > n$ (if not, then dimension of V will be larger than n). If this process ends when $k < n$, then $\text{span}(S) = \text{span}\{u_1, \dots, u_k\} = V$, and $\dim V = k < n$, which is also a contradiction. Thus $k = n$, and $\{u_1, \dots, u_k\} \subseteq S$ is a basis for V . $\square$

Let W be a subspace of V , recall that the quotient space V/W is the set

$$V/W = \{v + W \mid v \in V\}$$

with the vector addition and scalar multiplication defined by

$$(u + W) + (v + W) := (u + v) + W \text{ and } a \cdot (v + W) := av + W \text{ for all } a \in F.$$

We have shown that V/W with the operations forms a vector space structure. Intuitively, passing to V/W identifies vectors whose difference lies in W . Thus one might expect V/W to be generated by cosets represented by vectors complementary to W . This intuition is justified by the following proposition.

Proposition 7 (Exercise 1.6.35). *Let W be a subspace of V , and let $\{w_1, \dots, w_n\}$ be a basis for W , which extends to a basis $\{w_1, \dots, w_n, u_1, \dots, u_m\}$ for V , then*

$$\{u_1 + W, u_2 + W, \dots, u_m + W\}$$

is a basis for V/W . Therefore, $\dim V/W = \dim V - \dim W$.

Proof. Given any element of $x \in V/W$, then $x = v + W$ for some $v \in V$. Since v can be written as

$$v = a_1 w_1 + \dots + a_n w_n + b_1 u_1 + \dots + b_m u_m$$

for some $a_1, \dots, a_n, b_1, \dots, b_m \in F$, $v + W$ can be written as

$$\begin{aligned} v + W &= (a_1 w_1 + \dots + a_n w_n + b_1 u_1 + \dots + b_m u_m) + W \\ &= b_1 u_1 + \dots + b_m u_m + W \\ &= b_1(u_1 + W) + \dots + b_m(u_m + W) \end{aligned}$$

Therefore, $\{u_1 + W, u_2 + W, \dots, u_m + W\}$ is a generating set of V/W . It suffices to show that the set is linearly independent. Consider the relation

$$b_1(u_1 + W) + b_2(u_2 + W) + \dots + b_m(u_m + W) = 0_{V/W} = W$$

then we have

$$b_1 u_1 + b_2 u_2 + \dots + b_m u_m \in W,$$

so there exists $a_1, \dots, a_n \in F$ such that

$$b_1 u_1 + b_2 u_2 + \dots + b_m u_m = a_1 w_1 + \dots + a_n w_n.$$

That is, we have a relation

$$-(a_1 w_1 + \dots + a_n w_n) + b_1 u_1 + b_2 u_2 + \dots + b_m u_m = 0,$$

which forces that $a_1 = \dots = a_n = b_1 = \dots = b_m = 0$ since $\{w_1, \dots, w_n, u_1, \dots, u_m\}$ is a basis for V . $\square$

References

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